Incidence of Anterior Cruciate Ligament Tears and Reconstruction: A 21-Year Population-Based Study

Thomas L. Sanders, Hilal Maradit Kremers, Andrew J. Bryan, Dirk R. Larson, Diane L. Dahm, Bruce A. Levy, Michael J. Stuart and Aaron J. Krych

Am J Sports Med published online February 26, 2016

DOI: 10.1177/0363546516629944

AJSM PreView, published on February 26, 2016 as doi:10.1177/0363546516629944

Incidence of Anterior Cruciate Ligament Tears and Reconstruction

A 21-Year Population-Based Study

Thomas L. Sanders,*† MD, Hilal Maradit Kremers,†‡ MD, MSc, Andrew J. Bryan,† MD, Dirk R. Larson,‡ MS, Diane L. Dahm,† MD, Bruce A. Levy,† MD, Michael J. Stuart,† MD, and Aaron J. Krych,† MD
Investigation performed at Mayo Clinic, Rochester, Minnesota, USA

Background: The incidence of isolated anterior cruciate ligament (ACL) tears in the general population is not well defined.

Purpose/Hypothesis: The purpose of this study was to define the population-based incidence of ACL tears, describe trends in ACL injuries over time, and evaluate changes in the rate of surgical management. The hypothesis was that the incidence of ACL injury and the rate of subsequent ACL reconstruction increase over time.

Study Design: Cohort study; Level of evidence, 3.

Methods: The study population included 1841 individuals who were diagnosed with new-onset, isolated ACL tears (without concomitant ligament injury that required surgery) between January 1, 1990, and December 31, 2010. The complete medical records were reviewed to confirm diagnosis and to extract injury and treatment details. Age- and sex-specific incidence rates were calculated and adjusted to the 2010 US population. Poisson regression analyses were performed to examine incidence trends by age, sex, and calendar period.

Results: The overall age- and sex-adjusted annual incidence of ACL tears was 68.6 per 100,000 person-years. Incidence was significantly higher in male patients than in females (81.7 vs 55.3 per 100,000, $P < .001$). The incidence of isolated ACL tears decreased significantly over time in males ($P < .001$) but remained relatively stable in females. Age-specific patterns differed in male and female patients, with a peak in incidence (241.0 per 100,000) between 19 and 25 years in males and a peak in incidence (227.6 per 100,000) between 14 and 18 years in females. The rate of ACL reconstruction increased significantly over time in all age groups ($P < .001$).

Conclusion: With an annual incidence of 68.6 per 100,000 person-years, isolated ACL tears remain a common orthopaedic injury. Differences in age-specific incidence trends in male and female patients may potentially reflect differences in sports participation patterns through the high school and college years. The significant increase in the rate of ACL reconstruction over time may reflect changing surgical indications or an increasing desire among patients to return to high levels of activity after ACL injury.

Keywords: ACL tear; ACL reconstruction; epidemiology; incidence

Tears of the anterior cruciate ligament (ACL) are among the most frequently studied injuries in the orthopaedic literature. Despite this, the incidence of ACL tears in the general population has not been well defined but has been estimated to be between 30 and 78 per 100,000 people.[6,7,11,17,18,27] To date, the majority of studies describing the epidemiological patterns of ACL tears have focused on select athletic groups.[8,14,15,19] A recent meta-analysis reported incidence rates that varied by sex, sport, and participation in injury-reduction training programs.[21] While these studies provide valuable insight into the incidence of ACL injuries in select groups of athletes, they do not accurately reflect the injury burden in the general population.

Additionally, reports from large registry databases have shown an increasing incidence of ACL reconstruction surgery between 1994 and 2006.[4,12,13] These data could possibly be used as a surrogate to reflect an increasing incidence of ACL injury in the general population. However, it remains unclear whether the increase in surgical management is reflective of changes in injury patterns or simply changes in population incidence or operative indications over time.

To our knowledge, no studies have clearly established the population-based incidence of isolated ACL tears or concurrent changes in ACL injury and surgical reconstruction over time. To further evaluate this question, we report on our findings from a geographically defined population in which we retrospectively reviewed ACL injuries over a 21-

The American Journal of Sports Medicine, Vol. XX, No. X
DOI: 10.1177/0363546516629944

year period. Specifically, the purpose of this study is to (1) define the population-based incidence of ACL tears, (2) describe trends in ACL injuries over time, and (3) evaluate changes in the rate of surgical management. We hypothesize that both the incidence of ACL injury and the rate of subsequent ACL reconstruction are increasing over time.

METHODS

We performed a population-based historical cohort study using the resources of the Rochester Epidemiology Project (REP) in Olmsted County, Minnesota, which had a population of 144,260 in 2010. Briefly, the REP is a medical record linkage system that provides access to the complete medical records for all residents of Olmsted County, regardless of the medical facility in which the care was delivered.[22-25] Information is derived directly from physician-determined diagnostic codes and compiles complete diagnostic and procedural information from all medical centers in Olmsted County into one database. This population-based setting allows essentially complete ascertainment and follow-up of all clinically diagnosed cases of isolated ACL tears in a geographically defined community and provides the ability to access original medical records for confirmation of diagnosis and treatment. Due to the geographical isolation of Olmsted County from other large urban centers and availability of health care providers, most residents receive care within Olmsted County, which allows uninterrupted natural history studies.

We identified all subjects who were residents of Olmsted County, Minnesota, and had International Classification of Diseases, Ninth Revision (ICD-9) diagnosis codes consistent with acute ACL tears between January 1, 1990, and December 31, 2010. This search identified a total of 3494 potential subjects. The medical records of all subjects were reviewed in detail by the primary author (T.L.S.) to confirm the accuracy of diagnosis and gather data on injury type, laterality, associated meniscal injuries, magnetic resonance imaging (MRI) findings, and treatment details. Subjects were included if they had a new-onset and complete ACL tear in the study time period. Subjects were excluded if they had MRI or physical examination evidence of a partial tear of the ACL (either isolated anteromedial or posterolateral bundle tear), an associated posterior cruciate ligament tear at the time of ACL injury, fractures of the femur or tibia, a knee dislocation, an ACL tear that occurred 1 year or more before diagnosis (chronic ACL tear), or an ACL tear that occurred before 1990 but was first diagnosed during the study time period. *Isolated* was defined as not occurring with a concomitant ligament injury that required surgery; however, ACL tears with medial collateral ligament (MCL) sprains were included. ACL tears with concomitant meniscal or articular cartilage injury were also included. Upon medical record review, a total of 1841 new-onset, isolated ACL tears were identified and met the inclusion criteria as incident cases.

Age- and sex-specific rates of ACL injury were calculated by use of the number of new-onset, isolated ACL tears (incident cases) as the numerator and population estimates based on decennial census counts as the denominator, with linear interpolation between census years. Only patients who were residents of Olmsted County at the time of ACL injury and who fulfilled the study criteria were included in the incidence calculations. Overall incidence rates were age- and sex-adjusted to the 2010 white population of the United States. Ninety-five percent confidence intervals (95% CIs) for the incidence rates were constructed with the assumption that the number of incident cases per year follows a Poisson distribution. Incidence trends were examined by use of Poisson regression models with smoothing splines for age and calendar year (Figures 1 and 2). Similarly, for each isolated ACL incident case, the rate of ACL reconstruction within 1 year of injury was determined. Institutional review board approval was obtained for this study from all medical institutions in Olmsted County including Mayo Clinic and Olmsted Medical Center.

RESULTS

The final study cohort consisted of 1841 subjects with new-onset, isolated ACL tears; subjects had a mean $\pm$ SD age of 29.1 $\pm$ 11.0 years, and 59% were male (Table 1). A total of 1100 subjects (59.8%) were diagnosed with a meniscal tear at the time of presentation; 552 had medial tears (30.0%), 378 had lateral tears (20.5%), and 170 had both medial and lateral (9.2%) tears. Among patients diagnosed with complete ACL tears, MRI confirmed the tear in the majority (94.6%) of cases.

The overall age- and sex-adjusted annual incidence of isolated ACL tears was 68.6 (95% CI, 65.4-71.8) per 100,000 person-years. Male patients had a significantly ($P < .001$) higher annual incidence (81.7; 95% CI, 76.8-86.6 per 100,000) compared with female patients (55.3; 95% CI, 51.3-59.2 per 100,000). Table 2 and Figure 1 illustrate the annual incidence of ACL tears in different age groups, separately for male and female patients. Age-

*Address correspondence to Thomas L. Sanders, MD, Mayo Clinic, 200 First Street SW, Rochester, MN 55905 (email: sanders.thomas @mayo.edu).
†Departments of Orthopedic Surgery, Mayo Clinic, Rochester, Minnesota, USA.
‡Health Sciences Research, Mayo Clinic, Rochester, Minnesota, USA.

The findings and conclusions of this report are solely the responsibility of the authors and do not necessarily represent the official views of the National Institutes of Health.

One or more of the authors has declared the following potential conflict of interest or source of funding: This study was made possible by the Rochester Epidemiology Project, which is supported by the National Institute on Aging of the National Institutes of Health under award number R01AG034676. B.A.L. receives intellectual property (IP) royalties from and is a paid consultant for Arthrex Inc, receives research support from Biomet and Stryker, and receives IP royalties from VOT Solutions. M.J.S. receives IP royalties from and is a paid consultant for Arthrex Inc and receives research support from Stryker. D.L.D.'s spouse receives royalties from Tenex Health Inc. A.J.K. is a paid consultant for Arthrex Inc and receives research support from the Arthritis Foundation and Histogenics.

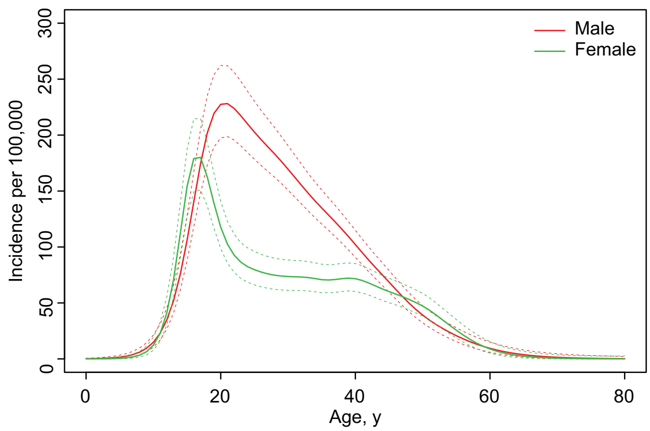

Figure 1. Age-specific incidence of anterior cruciate ligament (ACL) injury in males and females.

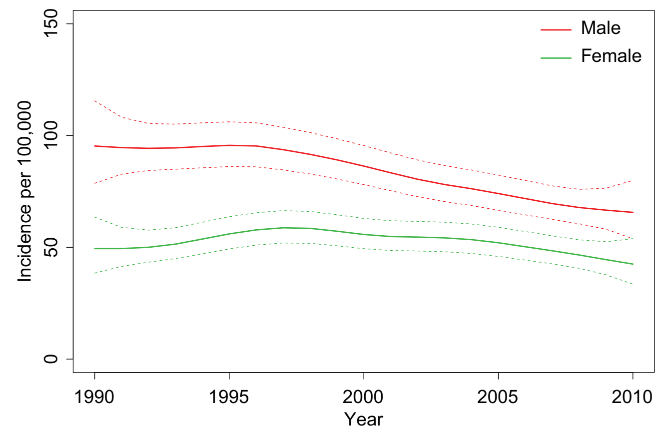

Figure 2. Trends in age-adjusted incidence of anterior cruciate ligament (ACL) tears in males and females.

TABLE 1
Characteristics of Incident Subjects With Anterior Cruciate Ligament Tears, 1990-2010

	Time Period				
	1990-1994	1995-1999	2000-2004	2005-2010	Total (N = 1841)
Age, mean ± SD, y	26.4 ± 8.7	29.3 ± 10.5	30.5 ± 11.5	29.8 ± 12.1	29.1 ± 11.0
Male sex, n (%)	257 (63.6)	278 (59.1)	271 (58.9)	290 (57.2)	1096 (59.5)
Meniscal injury, n (%)					
None	172 (42.6)	177 (37.7)	201 (43.7)	191 (37.7)	741 (40.2)
Medial	129 (31.9)	154 (32.8)	139 (30.2)	130 (25.6)	552 (30.0)
Lateral	66 (16.3)	93 (19.8)	84 (18.3)	135 (26.6)	378 (20.5)
Both	37 (9.2)	46 (9.8)	36 (7.8)	51 (10.1)	170 (9.2)
Magnetic resonance imaging confirmation, n (%)	366 (90.6)	443 (94.3)	437 (95.0)	496 (97.8)	1742 (94.6)

and sex-specific annual incidence in females was highest between 14 to 18 years (227.6 per 100,000) whereas among males, we observed a peak in incidence between 19 to 25 years of age (241.0 per 100,000).

Table 3 and Figure 2 illustrate the age- and sex-specific annual incidence of ACL tears over the 21-year time period between 1990 and 2010. We observed a significant decline in the incidence of ACL tears among males aged 14 to 18 and 26 to 35 years. In the 14- to 18-year age group, incidence declined from 270.0 per 100,000 in 1990-1994 to 184.9 per 100,000 in 2005-2010. Similarly, in the 26- to 35-year age group, incidence declined from 202.9 per 100,000 in 1990-1994 to 115.3 per 100,000 in 2005-2010. As a result of the decline in the incidence of ACL tears in these 2 age groups, the overall age-adjusted incidence in males declined from 96.0 per 100,000 in 1990-1994 to 70.9 per 100,000 in 2005-2010 ($P < .001$). In contrast, the incidence of ACL tears in female subjects remained relatively stable except among those aged 19 to 25 years. Incidence among 19- to 25-year-old females declined from 162.9 per 100,000 in 1990-1994 to 115.0 per 100,000 in 2005-2010 ($P = .004$).

Table 4 shows the rate of ACL reconstruction within 1 year of injury in 5-year time intervals between 1990 and 2010. We observed a significant increase in the rate of ACL reconstruction over the 21-year observation period in both men and women. Similarly, the rate of ACL reconstruction increased significantly in all ages between 14 and 45 years during the study period. Between 2005 and 2010, about 75% of subjects underwent ACL reconstruction within the first year of injury, including 98.3% of subjects younger than 18 years of age.

DISCUSSION

Despite the high frequency of patients with cruciate ligament injuries seeking care in orthopaedic clinics, the incidence of ACL rupture in the general population has not been well defined. This population-based study describes the incidence of isolated ACL tears and, for the first time, reports on trends in incidence of ACL injury and rate of surgical reconstruction in the general population over 2 decades.

We describe an overall ACL injury incidence of 68.6 per 100,000, which is similar to previous estimates based on national registry data.[18,20] Similar to the findings of Nordenvall et al,[18] our results indicate that the incidence of ACL injury in the general population is higher in males than females. In this study, the peak in ACL injury occurred in males aged 19 to 25 years, who may have been engaged in contact and/or high-energy sports as

TABLE 2
Age- and Sex-Specific Annual Incidence of Anterior Cruciate Ligament Injury, 1990-2010

	No. of Cases			Incidence Rate (per 100,000 person-years)		
	Female	Male	Total	Female	Male	Total
Age group						
≤13	3	8	11	1.1	2.9	2.0
14-18	203	190	393	227.6	203.0	215.0
19-25	132	267	399	113.2	241.0	175.4
26-35	172	349	521	81.1	166.1	123.4
36-45	142	205	347	69.4	101.7	85.4
≥46	93	77	170	20.7	20.0	20.4
Total (95% CI)	745	1096	1841	55.6 (51.7-59.8)	85.44 (80.5-90.7)	70.2 (67.1-73.5)
Total (95% CI)	745	1096	1841	55.3 (51.3-59.2)[a]	81.7 (76.8-86.6)[a]	68.6 (65.4-71.8)[b]

[a]Age-adjusted to 2010 US population.
[b]Age- and sex-adjusted to 2010 US population.

TABLE 3
Trends in Age- and Sex-Specific Incidence of Anterior Cruciate Ligament Injury, 1990-2010

	Time Period								
	1990-1994		1995-1999		2000-2004		2005-2010		
Age Group	No.	Incidence Rate	No.	Incidence Rate	No.	Incidence Rate	No.	Incidence Rate	*P* Value
Male									
≤13	3	4.8	1	1.5	2	3.0	2	2.3	.583
14-18	50	270.0	47	212.0	40	165.1	53	184.9	.022
19-25	60	242.0	65	252.5	70	261.8	72	214.9	.135
26-35	105	202.9	87	180.3	85	178.1	72	115.3	.002
36-45	32	76.7	62	125.0	53	100.7	58	100.5	.740
≥46	7	10.3	16	19.8	21	21.9	33	23.5	.064
Total	257	96.0	278	95.4	271	86.1	290	70.9	<.001
Female									
≤13	1	1.7	0	0.0	2	3.1	0	0.0	.643
14-18	43	239.7	55	261.7	40	175.1	65	237.5	.695
19-25	44	162.9	26	97.5	21	76.9	41	115.0	.004
26-35	37	70.1	55	113.3	48	100.6	32	50.8	.086
36-45	17	39.5	37	73.4	46	86.7	42	71.5	.057
≥46	5	6.1	19	20.0	32	28.8	37	23.1	.018
Total	147	52.1	192	63.3	189	57.9	217	50.9	.333

a predisposing factor for ACL injury. In fact, the incidence was higher among college-aged males (19-25 years) than among males of high school age (14-18 years). This finding is unexpected, as the majority of high school male athletes do not go on to participate in intercollegiate sports. However, this result may indicate that a large proportion of college-aged males continue to participate in intramural or club sports. Mountcastle et al[16] described a high rate of ACL injury among college-aged males participating in intramural rugby and football. Therefore, the high rate of ACL injury among college-aged males may reflect both ACL tears sustained in intercollegiate sports and also injuries due to recreational or intramural sports. The second largest subgroup of individuals who were at high risk for ACL injury were females aged 14 to 18 years, which is likely reflective of participation in competitive sports among high school females. Several studies have concluded that female athletes are at a high risk of ACL injury when participating in competitive sports.[1,2,10,16,20] As would be expected, there was a sharp decline in the rate of ACL injury in females aged 19 to 25 years, likely resulting from decreased participation in competitive sports after high school. Collectively, our results indicate that ACL tear is most common in females during the high school years but is most common in males during college or shortly thereafter.

During the study period, the incidence of ACL injury decreased significantly in male participants. Individuals ages 14 to 18 and 26 to 35 years experienced the greatest decrease in ACL injury. The reason for this decrease is unknown but could represent an increasingly sedentary lifestyle of males in these age groups. While the incidence of ACL tear in the general population of males decreased during this study, the rate of ACL injury among select groups of males (football players, soccer players, etc) might be increasing. In contrast, the incidence of ACL tears

TABLE 4
Trends in Anterior Cruciate Ligament Reconstruction Rates Within the First Year of Injury in Subjects With New-Onset Anterior Cruciate Ligament Tears, 1990-2010 (N = 1841)

	Time Period								
	1990-1994		1995-1999		2000-2004		2005-2010		
	n/N	%	n/N	%	n/N	%	n/N	%	*P* (Trend)
Age group, y									
≤13	1/4	25	0/1	0.0	3/4	75	2/2	100	.118
14-18	58/93	62.4	78/102	76.5	74/80	92.5	116/118	98.3	<.001
19-25	41/104	39.4	70/91	76.9	73/91	80.2	98/113	86.7	<.001
26-35	59/142	41.5	81/142	57.0	88/133	66.2	76/104	73.1	<.001
36-45	13/49	26.5	44/99	44.4	47/99	47.5	65/100	65	<.001
≥46	2/12	16.7	12/35	34.3	15/53	45.5	28/70	40	.058
Sex									
Male	122/257	47.5	169/278	60.8	184/271	67.9	223/290	76.9	<.001
Female	52/147	35.4	116/192	60.4	116/189	61.4	162/217	74.7	<.001

among females remained relatively stable during the study period. Despite this, female participants aged 19 to 25 years showed a significant decrease in the rate of ACL injury over time. As with males, the reason for this decrease is not completely understood but could reflect decreasing levels of activity among females who transition from high school athletics to a lower activity level in college. Previous studies have reported an increasing incidence of ACL reconstruction in the United States,[12,13] which many have thought to be a surrogate for an increasing incidence of ACL injury. However, based on the results of this study, the increasing incidence of ACL surgery is more likely related to treatment factors rather than to an increasing rate of cruciate ligament injury.

Our data indicate that the rate of ACL reconstruction in the general population is higher than previously reported rates. Nordenvall et al[18] described reconstruction rates of 30% within the first year of injury in Sweden. However, the rate of ACL reconstruction in our cohort was much higher during the first year after injury during a similar time period. The difference in ACL reconstruction rate could be related to differences in activity level or treatment patterns in the United States, which generally promote ACL reconstruction among physically active patients with subjective instability. During the study period, the rate of ACL reconstruction increased significantly for both men and women in nearly every age group. The cause of this trend is likely multifactorial and may represent an increasing activity level over time among patients and a greater desire to return to an active lifestyle after ACL injury. Similarly, the rising rate of ACL reconstruction could reflect changes in treatment pattern as surgeons develop better techniques and more predictable outcomes,[3,26] which may ultimately make providers more willing to offer ACL reconstruction over time. Furthermore, the recognition that ACL reconstruction can potentially reduce subsequent meniscal injury or progression to osteoarthritis might prompt surgeons to offer surgical reconstruction more frequently.[5,9]

Despite the population-based design and the 21-year observation period, our study has a number of potential limitations. First, the incidence cohort reflects only clinically diagnosed ACL tears and does not include individuals who did not seek medical care due to lack of symptom severity or lack of insurance. Similarly, MRI confirmation of injury was not available in some patients. In addition, the source population is primarily white, and the activity levels or findings in this cohort may not be representative of ethnically diverse populations or populations with different activity levels. Similarly, health care in this setting is provided by a relatively small number of providers, and the treatment patterns may not be representative of orthopaedic practices in other parts of the United States. Additionally, patient activity level was not recorded, which influences risk of exposure for ACL tears. Despite these, strengths of this study include the population-based design, chart-review based verification, determination of associated injuries, and the ability to account for all care provided in a geographically defined community. This study evaluated the rate of surgical reconstruction within 1 year of injury. Future studies should evaluate the incidence and predictive demographic variables of patients who choose nonoperative management or delayed (>1 year from injury) ACL reconstruction.

In conclusion, with an annual incidence of 68.6 per 100,000 person-years, isolated ACL tears remain a common orthopaedic injury. Secular trends indicate that the incidence of new-onset ACL tears is decreasing over time among males while remaining relatively stable in females. These trends are accompanied by a significant increase in ACL reconstruction over time that possibly reflects changing surgical indications and an increasing desire among patients to return to high levels of activity after ACL injury.

REFERENCES

1. Arendt E, Dick R. Knee injury patterns among men and women in collegiate basketball and soccer. *Am J Sports Med*. 1995;23(6):694-701.
2. Bjordal J, Arnoy F, Hannestad B, Strand T. Epidemiology of anterior cruciate ligament injuries in soccer. *Am J Sports Med*. 1997;25(3):341-345.

3. Carmont M, Scheffler S, Spalding T, Brown J, Sutton P. Anatomical single bundle anterior cruciate ligament reconstruction. *Curr Rev Musculoskelet Med*. 2011;4(2):65-72.
4. Csintalan R, Inacio M, Funahashi T. Incidence rate of anterior cruciate ligament reconstructions. *Perm J*. 2008;12(3):17-21.
5. Frobell R, Roos E, Roos H, Ranstam J, Lohmander L. A randomized trial of treatment for acute anterior cruciate ligament tears. *N Engl J Med*. 2010;363(4):331-342.
6. Gianotti S, Marshall S, Hume P, Bunt L. Incidence of anterior cruciate ligament injury and other knee ligament injuries: a national population-based study. *J Sci Med Sport*. 2009;12(6):622-627.
7. Granan L, Forssblad M, Lind M, Engebretsen L. The Scandinavian ACL registries 2004-2007: baseline epidemiology. *Acta Orthop*. 2009;80(5):563-567.
8. Gwinn D, Wilckens J, McDevitt E, Ross G, Kao T. The relative incidence of anterior cruciate ligament injury in men and women at the United States Naval Academy. *Am J Sports Med*. 2000;28(1):98-102.
9. Harkey M, Luc B, Golightly Y, et al. Osteoarthritis-related biomarkers following anterior cruciate ligament injury and reconstruction: a systematic review. *Osteoarthritis Cartilage*. 2015;23(1):1-12.
10. Hewett T, Ford K, Hoogenboom B, Myer G. Understanding and preventing ACL injuries: current biomechanical and epidemiologic considerations—update 2010. *N Am J Sports Phys Ther*. 2010;5(4):234-251.
11. Jannssen K, Orchard J, Driscoll T, van Mechelen W. High incidence and costs for anterior cruciate ligament reconstructions performed in Australia from 2003-2004 to 2007-2008: time for anterior cruciate ligament register by Scandinavian model? *Scand J Med Sci Sports*. 2012;22(4):495-501.
12. Lyman S, Koulouvaris P, Sherman S, Do H, Mandl L, Marx R. Epidemiology of anterior cruciate ligament reconstruction trends, readmissions, and subsequent knee surgery. *J Bone Joint Surg Am*. 2009;91:2321-2328.
13. Mall N, Chalmers P, Moric M, et al. Incidence and trends of anterior cruciate ligament reconstruction in the United States. *Am J Sports Med*. 2014;42(10):2363-2370.
14. Mihata L, Beutler A, Boden B. Comparing the incidence of anterior cruciate ligament injury in collegiate lacrosse, soccer, and basketball players: implications for anterior cruciate ligament mechanism and prevention. *Am J Sports Med*. 2006;34(6):899-904.
15. Moses B, Orchard J, Orchard J. Systematic review: annual incidence of ACL injury and surgery in various populations. *Res Sports Med*. 2012;20(3-4):157-179.
16. Mountcastle S, Posner M, Kragh J, Taylor D. Gender differences in anterior cruciate ligament injury vary with activity: epidemiology of anterior cruciate ligament injuries in a young, athletic population. *Am J Sports Med*. 2007;35(10):1635-1642.
17. Nielsen A, Yde J. Epidemiology of acute knee injuries: a prospective hospital investigation. *J Trauma*. 1991;31(12):1644-1648.
18. Nordenvall R, Bahmanyar S, Adami J, Stenros C, Wredmark T, Fellander-Tsai L. A population-based nationwide study of cruciate ligament injury in Sweden, 2001-2009: incidence, treatment, and sex differences. *Am J Sports Med*. 2012;40(8):1808-1813.
19. Owens B, Mountcastle S, Dunn W, DeBerardino T, Taylor D. Incidence of anterior cruciate ligament injury among active duty US military servicemen and servicewomen. *Mil Med*. 2007;172(1): 90-91.
20. Parkkari J, Pasanen K, Mattila V, Kannus P, Rimpela A. The risk for a cruciate ligament injury of the knee in adolescents and young adults: a population-based cohort study of 46,500 people with a 9 year follow-up. *Br J Sports Med*. 2008;42(6):422-426.
21. Prodromos C, Han Y, Rogowski J, Joyce B, Shi K. A meta-analysis of the incidence of anterior cruciate ligament tears as a function of gender, sport, and a knee injury-reduction regimen. *Arthroscopy*. 2007;23(12):1320-1325.
22. Rocca WA, Yawn BP, St Sauver JL, Grossardt BR, Melton LJ. History of the Rochester Epidemiology Project: half a century of medical records linkage in a US Population. *Mayo Clin Proc*. 2012;87(12): 1202-1213.
23. St Sauver JL, Grossardt BR, Leibson CL, Yawn BP, Melton LJ, Rocca WA. Generalizability of epidemiological findings and public health decisions: an illustration from the Rochester Epidemiology Project. *Mayo Clin Proc*. 2012;87(2):151-160.
24. St Sauver JL, Grossardt BR, Yawn BP, et al. Data resource profile: the Rochester Epidemiology Project (REP) medical records-linkage system. *Int J Epidemiol*. 2012;41(6):1614-1624.
25. St Sauver JL, Grossardt BR, Yawn BP, Melton LJ, Rocca WA. Use of a medical records linkage system to enumerate a dynamic population over time: the Rochester Epidemiology Project. *Am J Epidemiol*. 2011;173(9):1059-1068.
26. Van Eck C, Lesniak B, Schreiber V, Fu F. Anatomic single- and double-bundle anterior cruciate ligament reconstruction flowchart. *Arthroscopy*. 2010;26(2):258-268.
27. Webb J, Corry I. Epidemiology of knee injuries: diagnosis and triage. *Br J Sports Med*. 2000;34:227-228.